RETURN & RISK ANALYSIS

BAJAJ FINANCE LTD. vs NIFTY 50

Six-Month Analysis Using Daily Market Data

Study Period
10 February 2026 – 10 August 2026

Company: Bajaj Finance Ltd.
Market Benchmark: NIFTY 50
Data Frequency: Daily
Observations: 121

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1. Introduction

This report presents a comparative return and risk analysis of Bajaj Finance Ltd. against the NIFTY 50 index over a six-month period from 10 February 2026 to 10 August 2026. Daily closing-price data was obtained from NSE India and processed using Python.

The analysis focuses on daily logarithmic returns, average daily return, daily volatility, and maximum drawdown. These measures are used to evaluate the return generated by Bajaj Finance and the level of risk associated with it relative to the broader Indian equity market represented by the NIFTY 50.

The main objective is to determine whether Bajaj Finance was more volatile than the market during the selected period and to assess whether its observed risk-return characteristics support a defensive-stock interpretation.

2. Objectives

- Obtain six months of daily closing prices for Bajaj Finance and the NIFTY 50.
- Calculate daily logarithmic returns for both series.
- Compute average daily return and daily volatility.
- Measure maximum drawdown over the study window.
- Compare Bajaj Finance's return and risk characteristics with the NIFTY 50.
- Assess whether Bajaj Finance displayed defensive characteristics during the selected period.

3. Data Used

Particular	Bajaj Finance	NIFTY 50
Security Type	Equity Share	Market Index
Symbol	BAJFINANCE	NIFTY 50
Data Source	NSE India	NSE India
Study Period	10-Feb-2026 to 10-Aug-2026	10-Feb-2026 to 10-Aug-2026
Frequency	Daily	Daily
Observations	121	121
Variable Used	Closing Price	Closing Price

Both datasets were sorted into chronological order before the calculations. The analysis uses the common six-month trading window so that the return and risk measures are directly comparable.

4. Methodology

4.1 Daily Log Return

For each trading day t, the daily logarithmic return is calculated as:

$$r_t = \ln(P_t / P_{t-1})$$

where P_t is the closing price on day t and P_{t-1} is the closing price on the previous trading day. Log returns are useful because they are additive over time and are commonly used in return and volatility analysis.

4.2 Average Daily Return

Average daily log return is the arithmetic mean of the daily log-return series:

$$\text{Average Return} = \Sigma r_t / n$$

4.3 Daily Volatility

Daily volatility is measured as the sample standard deviation of daily log returns:

$$\sigma = \sqrt{[\ \Sigma(r_t - \bar{r})^2 / (n - 1)\ ]}$$

A higher standard deviation indicates greater day-to-day variability in returns.

4.4 Maximum Drawdown

Maximum drawdown measures the largest peak-to-trough decline during the study period. A running peak is maintained and each day's price is compared with that peak:

$$\text{Drawdown}_t = P_t / \text{RunningPeak}_t - 1$$

Maximum Drawdown = minimum of all daily drawdowns

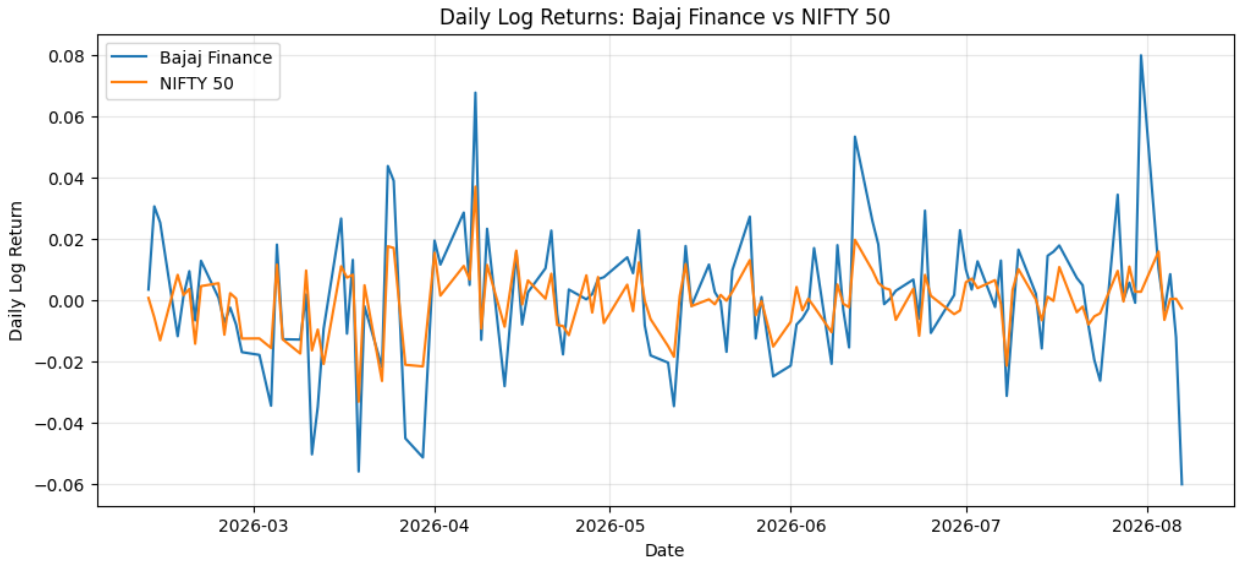
5. Matched-Window Statistics

Metric	Bajaj Finance	NIFTY 50
Average Daily Log Return	0.0918%	-0.0450%
Daily Volatility	2.2080%	1.0525%
Maximum Drawdown	-22.26%	-13.96%

Volatility ratio (Bajaj Finance / NIFTY 50): 2.10×

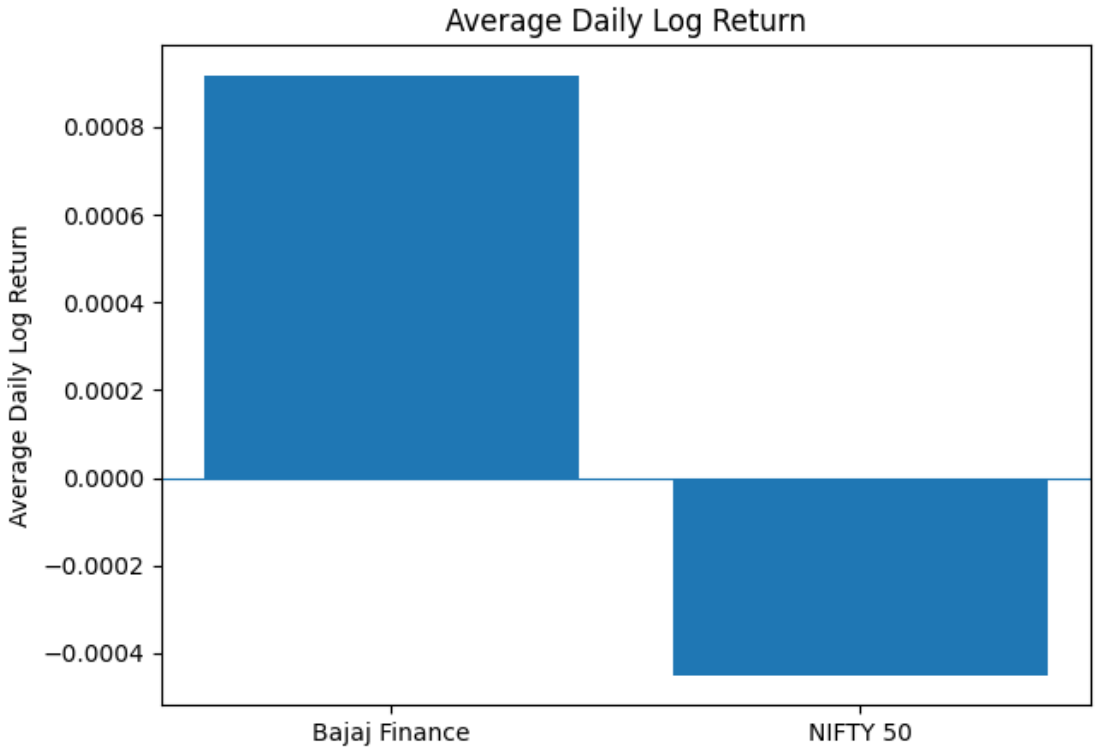
6. Graphical Analysis

Figure 1. Daily Log Returns: Bajaj Finance vs NIFTY 50



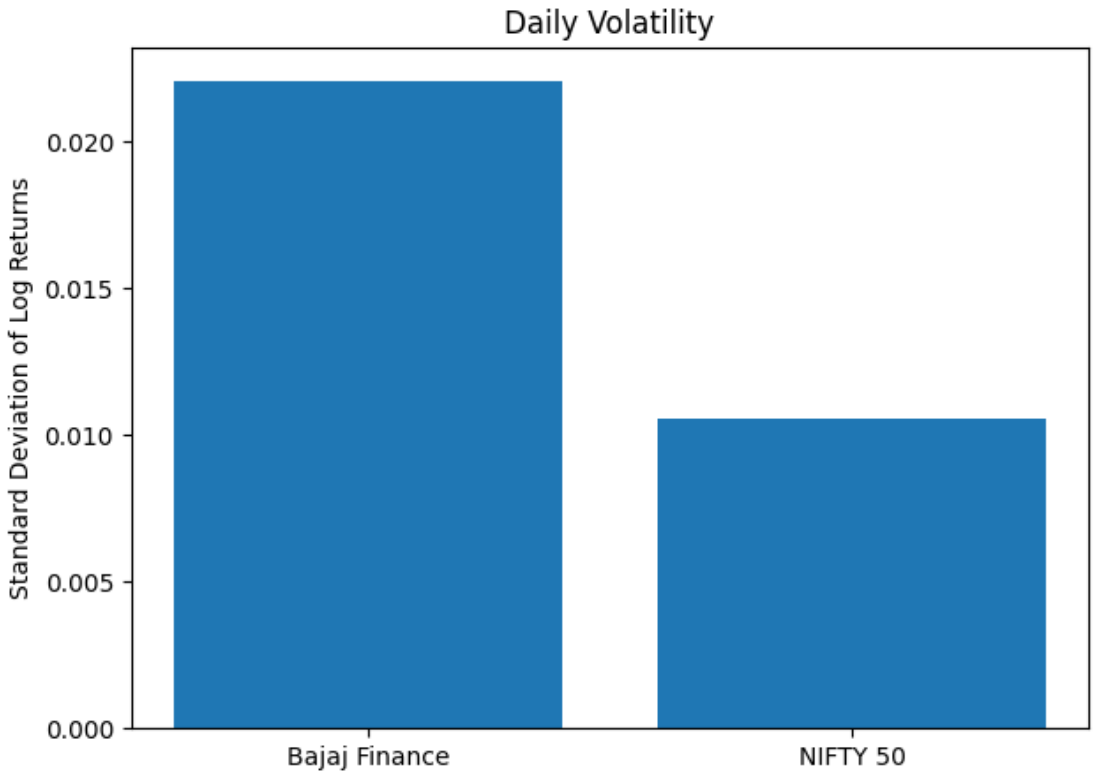
The daily log-return series shows noticeably larger positive and negative movements for Bajaj Finance than for the NIFTY 50. The wider swings in the Bajaj Finance series are consistent with its higher measured daily volatility.

Figure 2. Average Daily Log Return



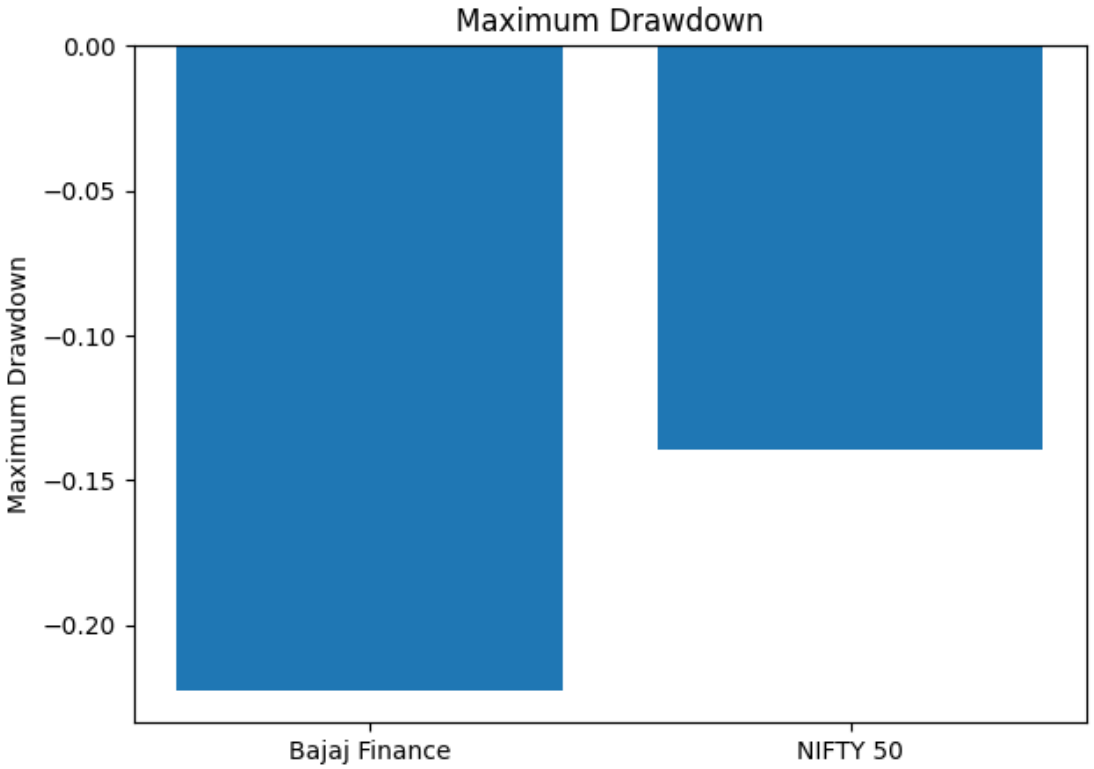
Bajaj Finance recorded a positive average daily log return of 0.0918%, while the NIFTY 50 recorded -0.0450%. Thus, Bajaj Finance generated the higher average daily return over this study window.

Figure 3. Daily Volatility



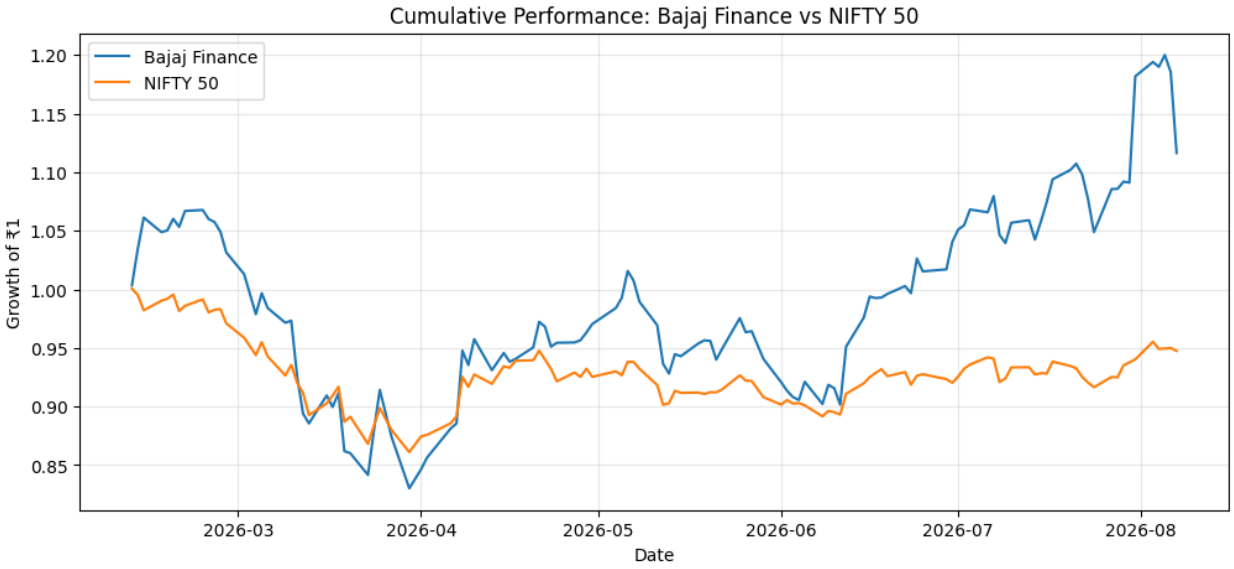
Bajaj Finance's daily volatility was 2.2080%, more than twice the NIFTY 50 volatility of 1.0525%. This indicates substantially greater day-to-day return fluctuation in Bajaj Finance during the period.

Figure 4. Maximum Drawdown



Bajaj Finance experienced a maximum drawdown of -22.26%, compared with -13.96% for the NIFTY 50. Therefore, the stock experienced a deeper peak-to-trough decline during the selected period.

Figure 5. Cumulative Performance: Bajaj Finance vs NIFTY 50



The cumulative-performance chart shows that Bajaj Finance finished the period above its initial normalized value and substantially above the NIFTY 50 by the end of the displayed window, although its path included materially larger intermediate fluctuations.

7. Interpretation and Comparison

7.1 Was Bajaj Finance More Volatile Than the Market?

Yes. Bajaj Finance recorded daily volatility of 2.2080%, while the NIFTY 50 recorded 1.0525%. The volatility ratio was approximately 2.10×, meaning Bajaj Finance's daily return fluctuations were more than twice those of the market benchmark during the study period.

7.2 Return Comparison

Bajaj Finance produced a positive average daily log return of 0.0918%, whereas the NIFTY 50 produced a negative average daily log return of -0.0450%. Thus, Bajaj Finance offered the stronger average daily return over this particular six-month window.

7.3 Downside Risk Comparison

The maximum drawdown result indicates greater downside exposure for Bajaj Finance. Its -22.26% maximum drawdown was substantially deeper than the NIFTY 50's -13.96% drawdown. This suggests that investors in Bajaj Finance would have experienced a larger peak-to-trough loss during the worst part of the selected period.

8. Defensive Stock Assessment

A defensive stock is generally expected to be relatively less sensitive to broad market movements and to provide greater resilience during market stress. In quantitative terms, beta below 1 is a standard indicator of lower market sensitivity. Volatility and drawdown provide complementary evidence about absolute risk and downside behavior.

Based on the statistics calculated in this report, Bajaj Finance does not show strong defensive characteristics during the selected six-month period. Its daily volatility was approximately 2.10 times the NIFTY 50 volatility, and its maximum drawdown was materially larger than that of the market.

However, the present analysis does not calculate beta or correlation. Therefore, the defensive-stock classification should be stated cautiously: the observed volatility and drawdown do not support a defensive interpretation, but a formal beta-based conclusion would require an additional regression/covariance analysis using the paired daily return series.

9. Conclusion

The six-month return and risk analysis shows that Bajaj Finance generated a higher average daily log return than the NIFTY 50, with values of 0.0918% and -0.0450%, respectively. However, this higher return was accompanied by considerably greater risk.

Bajaj Finance's daily volatility of 2.2080% was approximately 2.10 times the NIFTY 50 volatility of 1.0525%. Its maximum drawdown of -22.26% was also larger than the market's -13.96% drawdown.

Overall, Bajaj Finance delivered stronger average daily returns during the selected period but exhibited substantially higher variability and downside risk. Consequently, based on the observed volatility and drawdown, the stock does not appear to have behaved as a defensive stock during this particular six-month period. This conclusion is specific to the study window and should not be generalized to all market periods.

10. References

1. National Stock Exchange of India (NSE India), historical equity and index market data.
2. Bajaj Finance Ltd. (BAJFINANCE), daily closing-price data for the study period.
3. NIFTY 50, daily closing-price data for the study period.
4. Tutorial 02 – Detailed Solution Methodology: Return, Volatility and Drawdown Analysis.